

VCF-RDFizer: source-checkable semantic conversion for linked, policy-aware genomic querying

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Abstract

Background: The Variant Call Format (VCF) is widely used, compact, and readily parsable by a variety of tools, but two key challenges remain for its wider use in clinical contexts. (1) Information crucial for interpretation is not represented explicitly within the format, which makes integrating VCF data with other data sources such as personal, clinical, or experimental data difficult. (2) Privacy and consent information is represented at file level and often not in a machine-readable way. Here we demonstrate the representation of VCF in the Resource Description Framework (RDF) to address these challenges, and present VCF-RDFizer, a command-line tool that mediates the conversion of VCF into source-checkable RDF using the public VCF Core vocabulary.

Results: We evaluate VCF-RDFizer using public genomic datasets, source-based validation, scaling experiments, and a linked workflow combining genomic observations with external knowledge and simulated access policies. The results show that VCF data can be converted into an explicit RDF representation while retaining the information needed to check the transformed data against its source. Complementary query- and constraint-based validation expose different classes of transformation errors, demonstrating the value of validating both data content and graph structure. The resulting RDF representations also support queries that combine genomic observations with external variant knowledge, population information, and requester-specific policies while reproducing the results of an equivalent conventional VCF workflow. These capabilities come with additional

conversion and indexing costs: semantic processing is therefore most advantageous for workflows that repeatedly reuse, integrate, or govern genomic data, whereas native VCF tools remain better suited to many direct or narrowly targeted analyses.

Conclusions: VCF-RDFizer supports checkable conversion and reusable, policy-aware research queries; it is not a universal replacement for VCF. Its value depends on the relationships needed, the validation coverage achievable, and the cost of conversion and indexing.

Keywords: Genomic variation, Variant Call Format, RDF, SPARQL, Semantic validation, Data-use policies

1 Background

The Variant Call Format (VCF) is widely used to represent and exchange genomic variation data in research and clinical sequencing, including in pharmacogenomics and precision medicine use cases [1–3]. Variant data are increasingly interpreted alongside external biomedical resources, sample metadata, and governance constraints. This integration of data establishes links between variants and external knowledge, but VCF does not support representing these links in a queryable or machine-interpretable way. Additionally, genomic data are highly sensitive personal data, and access may need to be controlled per region or per variant rather than per file, which VCF does not support. Finally, questions over VCF files are usually answered with command-line utilities such as bcftools [4] or a custom pipeline, rather than by a declarative query over a shared model.

Consider a question that combines a participant’s variants, ClinVar classifications, population allele frequencies, and the rules governing a requester’s access. Conventional tools can answer it. However, the workflow must make its assumptions about variant identity, annotation meaning and permitted release explicit somewhere. Semantic-genomics initiatives already connect variant observations to curated resources and governance frameworks [5, 6]. The question here is whether a reusable, checkable representation can support that work without hiding the cost of obtaining it.

The Resource Description Framework (RDF) represents relationships as triples whose resources can be identified by Internationalized Resource Identifiers (IRIs) [7, 8]. Linked Data practice uses those identifiers to connect independently managed resources [9–12]; SPARQL queries their relationships [13]; and the Open Digital Rights Language (ODRL) expresses policies about their use [14]. Neither RDF nor a policy annotation enforces privacy by itself. A policy-aware workflow also needs an evaluator, a defined release boundary, and deployment controls.

Existing VCF-to-RDF routes include TogoVar, Jvarkit, SPARQLing Genomics, and BioInterchange [5, 15–18]. They differ in their target models and treatment of file metadata and sample data; the wider documentation-based comparison is in Supplementary Table S5. Native frameworks such as Hail, GEMINI and TileDB-VCF

also integrate variant annotations [19–21], while Beacon, htsget, and FHIR Genomics support discovery, retrieval, and clinical exchange [22–24]. These are alternatives for particular tasks, not systems that require replacement by RDF.

Biomedical semantic tools provide useful precedents for combining transformation with assessment: SPHN Schema Forge generates schemas and validation artifacts, FAIR-Checker combines queries and shapes to assess metadata, and a FHIR RDF enrichment pipeline evaluates the information needed by its intended workflow [25–27]. For VCF, the corresponding challenge is to preserve source-dependent information, test the resulting representation, and demonstrate an integration task at a measured computational cost.

We present VCF-RDFizer, a conversion and validation workflow that utilizes the public VCF Core vocabulary [28]. Its contribution is the evaluated combination of source-checkable conversion, separately maintained links and policy-aware release, rather than a new claim that VCF can be converted to RDF. Three questions organize the evaluation: **RQ1**, which aspects of VCF information are preserved and checkable; **RQ2**, what the explicit representation enables in a linked, policy-aware workflow; and **RQ3**, under which workloads and representation choices its costs are practical. The vocabulary is developed separately [29]; the representation decisions needed to assess the tool are specified here.

2 Implementation

2.1 Representation contract

VCF-RDFizer represents the source document, not only an interpreted variant. Its VCF Core output separates file and header assertions, variant records, and sample/genotype assertions (Table 1). The vocabulary aligns with FALDO for coordinates, the Sequence Ontology for variant types, HERO for genomic constructs, and features of the GA4GH Variation Representation Specification (VRS) [30–34]. These alignments connect VCF-specific structure to existing models; they do not replace the need to preserve the source’s interpretation context.

Deterministic IRI templates identify resources by source file and record position. The same biological alteration in two files therefore remains two observations, each with its own quality values, annotations and provenance. A separate linkset can connect both to a shared variant identifier (Figure 1). File-scoped observation identity and cross-file variant identity serve different purposes and are not collapsed.

Sample data have two profiles. *Expanded* makes sample calls and their FORMAT values directly addressable graph resources. *Condensed* stores sample-ordered vectors, retaining the values but requiring their positions to be decoded. Both retain file, header and record context. For V records, S samples and F FORMAT keys, the repeated sample structure scales as $\mathcal{O}(S + VSF)$ triples expanded and $\mathcal{O}(S + VF)$ condensed, assuming bounded ploidy and value arity. Condensed still stores the VSF values inside literals: fewer triples do not mean that the sample information disappears. The detailed model and derivation are in Supplementary Section S3.

Table 1 Representation contract and assessment boundary. The complete capability matrix is in Supplementary Table S6; Q1–Q13 are specified in Supplementary Table S7, and shapes are the Shapes Constraint Language (SHACL) constraints of Section 2.2.

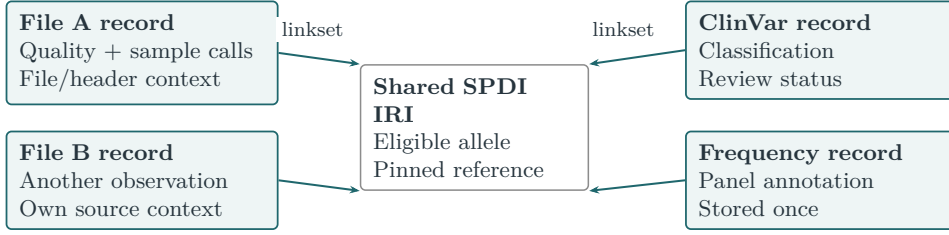
Information	Representation	Assessment and limit
File and header	Versioned file, declarations and metadata linked to their values	Q7–Q10 and shapes; header attribute-value agreement needs the deeper shapes.
Record and alleles	Core fields, ordered alleles, typed values and explicit missingness	Q1–Q4, Q11 and shapes; the digest is a summary, not exhaustive equality.
INFO	Raw string plus optional typed, structured values	Q12 and structural checks; preservation differs from interpreting every annotation.
Samples and FORMAT	Expanded sample-call resources or condensed, sample-ordered vectors	Q5–Q6 and Q13; individual condensed values require decoding.
Version-dependent constructs	VCF 4.1–4.5 file classes; symbolic and newer fields carried through	Fixture coverage, not full interpretation of all constructs; local-to-global allele resolution is not implemented.
Links and policies	Separate linksets and policy-defined release views	Evaluated independently of conversion; identity assumptions and simulated policies bound the use case.

2.2 Conversion, validation and extension points

The Python command-line tool separates conversion, optional representation construction, validation and packaging. Record, header and file metadata become tab-separated logical sources for RDF Mapping Language (RML) mappings, executed by RMLStreamer [35, 36]. Direct emitters supply the remaining typed fields and sample layers. This is a hybrid implementation: RML produces the nine-triple record backbone, 5.3% of the HG005 graph, while emitters produce the remaining 94.7%. Editing mappings alone therefore does not change the complete representation.

Parts are appended to plain or gzip-framed N-Triples and deleted after appending. Optional Header–Dictionary–Triples (HDT) and Parquet-based COTTAS outputs are built incrementally, without keeping a full decompressed duplicate [37, 38]; gzip and Brotli packages are separate transport choices [39]. Before source RDF is removed, native readers decode each selected representation and compare its triple count with the source. This checks decoding and counts, not all values. QLever builds its own query index; its measurements must be distinguished from engines that query HDT or COTTAS directly [40–42]. Construction details are in Supplementary Section S3.2.

A. Keep source records separate; add shared allele identity



Schematic resources. Shared identity supports joins under the stated normalization assumptions; source records are not merged.

B. Separate conversion, linked reuse and operational cost

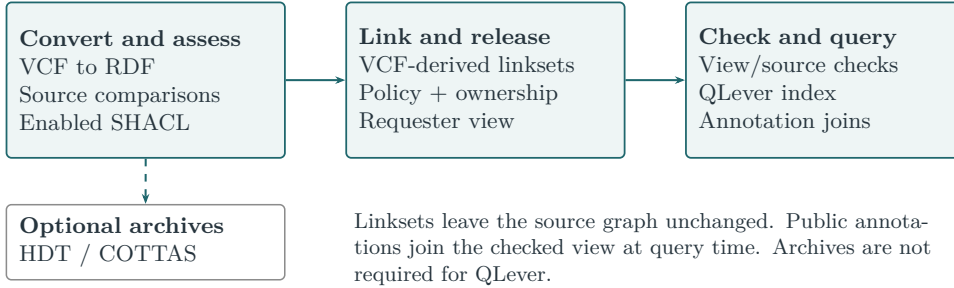


Fig. 1 Source observations remain separate while their links can be reused. Conversion creates file-scoped records with header and sample context. Separate linksets connect eligible alleles to a shared identifier under the stated reference and normalization assumptions. The evaluated workflow joins these links to ClinVar and, for the cohort, a separate frequency file. Policies select a requester's release view; source and policy checks assess that view before querying. Optional archives are a separate branch, not a prerequisite for QLever.

Readable output is not necessarily faithful output. The validation layer therefore pairs thirteen SPARQL queries with answers derived from the VCF by cyvcf2, using bcftools for exact FILTER text [4, 43, 44]. Q1–Q6 summarize records, alleles and genotypes; Q7–Q10 assess metadata and graph inventories; Q11–Q13 compare IRI-bound value summaries. The last three count values in up to 256 buckets keyed on the first byte of SHA-256, not collision-free, value-by-value comparisons. Fourteen preflight queries and SHACL constraints provide additional checks [45]. The default shapes assess cardinality and datatypes; two deeper profiles assess uniqueness and value agreement. Their execution limits and measured sensitivity are reported separately.

Linking leaves the converted graph unchanged. The three linker tiers rewrite existing identifiers, join digest-pinned reference data, or confirm identifiers through a rate-limited live service. The evaluated SPDI (Sequence Position Deletion Insertion) linker minimally trims explicit alleles and harmonizes contig names to RefSeq accessions [46]; a second linker records Ensembl gene overlaps. This establishes equality joins under the tested normalization assumptions, not canonical identity across all equivalent variant expressions.

The policy plug-in evaluates ODRL rules targeting files or selections declared in SPARQL. Its profile implements default-deny and deny-overrides, with purposes drawn from a subset of the Data Use Ontology (DUO) [14, 47]. An ownership rule states what withholding a resource also removes: a record’s dependent calls and alleles, or a file’s header. Each release view is checked for policy-digest agreement, permitted membership, absence of dangling references, and record equality with a VCF-text-derived oracle. This is policy-aware release, not anonymization or a complete security deployment; obligations are recorded but not enforced.

2.3 Evaluation methods

2.3.1 RQ1: preservation and validation

The public corpus spans five Personal Genome Project donations, an HG002 precisionFDA/Genome in a Bottle (GIAB) challenge file, the 2,504-sample 1000 Genomes chromosome-20 call set, HGSVC2 structural variants, and the HG004 and HG005 GIAB benchmarks [48–52]. The corpus experiment processes the first 250,000 records of eight inputs and HG005 whole; the large cohort is excluded from the expanded-profile corpus run and assessed through controlled slices. The inputs are listed in Supplementary Table S3.

Paired validation covers the difficult-input fixtures, a 10,000-line (9,986-record) single-sample fixture under both sample profiles and three RDF artifacts, and slices up to 100,000 HG005 records. A separate follow-up tests 250,000 records from the consumer genome NG131FQA1I. This run is reported separately because it ran after the campaign, with a later build, on QLever without shapes. A fault-injection experiment tests 113 deliberate corruptions with queries alone, with the default shapes, and with all shape profiles; unmutated controls test whether the validator rejects valid inputs. Separate retrieval runs compare the same thirteen answers at whole-genome scale without SHACL. The specification, exclusions and oracle dependencies are in Supplementary Section S4.

2.3.2 RQ2: linked, policy-aware querying

The application question is: *which participant-variant-gene matches connect an observed allele to a ClinVar classification in the fixed 81-gene American College of Medical Genetics and Genomics secondary-findings list (ACMG SF v3.2), and which may each requester receive?* The benchmark uses this frozen gene list, not a claim about the latest recommendations [53]. It accepts ClinVar records with at least one review star and includes all admitted classifications. Pathogenic/likely pathogenic (P/LP) matches are counted separately; the workflow does not implement the additional criteria for reporting clinical secondary findings.

The RDF route links converted genomes to ClinVar and gene resources, materializes requester views, and queries each view with SPARQL. The baseline harmonizes contigs, annotates variants with bcftools, and evaluates the same case definition in a separate Python implementation. Sorted result tuples must agree by participant, gene, contig, position and alleles before timings are compared. Counts are matches, not

unique participants. Shared preprocessing and case definitions limit the independence of the comparison even though the downstream implementations are separate.

Three arms test integration, breadth and scale: five genomes from three sources; 104 unrelated high-coverage 1000 Genomes participants, four per population [54]; and HG005 whole. The first two use normalized, split records within the gene spans. The cohort removes repeated panel-level INFO from each participant and represents the required frequencies once in a separate sites-only graph. The whole-genome answer is also checked against HG005’s restricted-input answer. ClinVar is fixed at its 2026-09-13 release [55].

All consents are simulated. Requesters represent clinical care, disease-specific cardiovascular research and general research. File-level permissions, withdrawals and a rule restricting records in 28 cancer-predisposition genes determine release. Each participant is represented by a single-sample file; this does not test selective masking within a multi-sample cohort file. Four applied rule changes compare where configuration or execution code must change, not user performance or maintainability. Supplementary Section S5 gives the complete protocol.

2.3.3 RQ3: operational trade-offs

Record ladders use 10,000, 100,000 and one million HG005 records, with three replicates, plus the whole genome. Sample ladders hold 10,000 cohort records and INFO fixed while varying one to 2,504 samples under both profiles. We measure stage timings, resident memory, transient workspace and retained artifact sizes separately. Whole-corpus measurements are single runs, not replicated estimates.

Retrieval compares individual questions, a thirteen-question batch, and five regional questions against scanned and tabix-indexed VCF access. Regional windows span 1 kb–10 Mb, with twenty windows per size and three replicates; indexed readers discard overlapping records whose POS lies outside the window so that all routes answer the same question. Setup is separate from query time. The principal machines have eight cores and 33.6 GB RAM, with no concurrent benchmark jobs. The 143-cell base campaign uses v3.1.0; the use case converts with v3.2.0 and links and governs with pre-release plug-in builds; later retrieval, review-validation and plug-in runs are separate experiment families. Supplementary Section S1 records versions, hosts, exclusions and unresolved archival details rather than treating them as one release.

3 Results

3.1 RQ1: preservation is checkable, but validation coverage is uneven

All 984 applicable paired comparisons in the base campaign agreed; the two genotype questions on each of the zero-record fixture’s two artifacts were not applicable. All 1,144 cross-check invariants held. Where all four engines were used, 25 of 25 validations agreed; the other 51 completed validations were single-engine runs. Native decoding preserved counts in all 152 HDT/COTTAS artifacts. These findings answer different questions and are separated in Table 2.

Table 2 Validation evidence by scope. Follow-up runs are not added to the base campaign’s denominator. A query pass establishes equality of the specified summaries; a shape pass establishes the enabled constraints. Neither proves complete semantic equivalence.

Assessment	Observed result	Scope or limit
Base paired comparisons	984/984 applicable equal; 4 not applicable	Fixtures and slices to 17.1M triples; 25/25 four-engine validations agreed.
Native artifact checks	152/152 counts equal	118 HDT and 34 COTTAS artifacts; decoding and counts only.
Separate HG005 retrieval	All thirteen answers equal	To 657.4M triples; no SHACL, outside the base denominator.
Consumer-genome follow-up	10/13 summaries equal	58.2M triples; three oracle/normalization mismatches, with direct record-field agreement.
Injected faults: queries	96/113 detected	Defined fault set on small graphs.
Queries + default shapes	96/113 detected	No additional faults detected in this set.
Queries + all shapes	113/113 detected	Both unmutated controls pass; costly opt-in profiles.

The consumer-genome follow-up exposed a limit of the validator rather than a demonstrated conversion defect. On 250,000 NG131FQA1I records (58.2M triples), Q9 and Q10 disagreed because their expected inventories omitted 30,910 emitted phase sets. Q11 disagreed because QLever canonicalized decimal quality scores (QUAL): 1,998 records with trailing zeros hashed differently. Direct comparison found the record fields equal to the VCF text for all 250,000 records, and canonicalizing the oracle’s QUAL values reproduced QLever’s buckets. This is supporting diagnostic evidence, not a passing end-to-end validation run. The initial SHACL attempt exhausted memory; the completed comparison omitted shapes.

Fault injection explains why both validation layers matter. Queries detected 96 of 113 corruptions. The remaining 17, spanning ten classes, affected relationships or values the summaries did not inspect, including allele references, sample indices, phasing and header attributes. The default shapes detected none of these additional faults; all three profiles together detected all 113. The two deeper profiles took 33.7 and 73.2s on a roughly 2,000-triple graph, compared with 1.7s for the default. Their coverage cannot therefore be assumed for routine large-file validation. Default-profile checks on 58 ordinary validations found no violations, but that is a conformance result, not a fault-detection rate.

Testing also found and resolved three substantive defects: missing header-declared samples in an empty file; genotype references to allele resources omitted in a raw-INFO configuration; and a withheld file’s header surviving in release views. Record counts alone could not show any of them. The findings motivate checks of dependencies and ownership as well as totals. Version fixtures (VCFv4.1–4.5 and an undeclared version)

converted, and nine of eleven difficult inputs converted and validated; a truncated gzip was refused, while a malformed sites-only fixture left that paired-coverage test incomplete. Supplementary Sections S2, S4 and S8 retain these outcomes, repeatability checks and unresolved oracle limitations.

3.2 RQ2: shared identifiers support reusable joins and checked release

The RDF and bcftools routes returned identical requester-specific match sets in every arm (Table 3). Each release view passed the policy and source checks. HG005’s whole-genome answer matched its restricted-input answer exactly: 1,496 participant–variant–gene tuples, not 1,496 people. The unrestricted P/LP subset contained zero matches in the five-genome and whole-genome arms and eight in the cohort. These are classifications retrieved under the case definition, not clinically adjudicated secondary findings.

Table 3 Identical match sets from the RDF and bcftools routes. Counts are participant–variant–gene tuples. CC denotes clinical care, DS disease-specific cardiovascular research, and GRU general research. “All” excludes policy filtering. Graph sizes contain genomic observations before linksets; the cohort’s rare subset additionally joins the separate frequency graph at panel allele frequency (AF) < 0.01.

Input / graph	Files	All	CC	DS	GRU
Five genomes / 9.0M	5	7,211	2,878	2,988	1,987
Cohort / 67.8M	104	173,102	63,529	92,870	48,207
Rare subset		4,154	1,621	2,198	1,144
Whole HG005 / 662.4M	1	1,496	1,496	0	0

The cohort’s frequency query is a concrete benefit of keeping shared information separate. A 68,635-site graph (6.5M triples) replaced approximately 857M triples of panel INFO otherwise repeated across participant graphs. The SPDI links then joined genomic observations, ClinVar and panel frequencies without embedding those frequencies in each genome. The baseline’s position-and-allele lookup returned the same rare-variant matches. The reduction comes from projecting and factoring the input information, not from RDF compression.

Shared identity also changed where one release rule was expressed. Restricting HFE C282Y to clinical care required a policy edit without modifying the RDF policy engine; the baseline required a small code and data edit. On the change fixture, the identity-targeted rule also selected HG002’s record written on contig 6 rather than chr6. Three other changes were similarly small data edits in both routes. This demonstrates reusable configuration for the tested change, not a general advantage in usability or lines of code. Full change-impact and linking results are in Supplementary Section S5.

The workflow remains costly (Table 4). Converting the whole genome and ClinVar to RDF took 54 min before linking, release construction, checking or query indexing. The clinical view released the complete genome, while the other two views were empty; this is scale evidence for that policy pattern, not a selective whole-genome masking test. The run peaked at 13.1 GB memory and 34 GB disk. Similar query times across arms suggest that the fixed ClinVar join dominated this workload; they do not establish scale-independent SPARQL performance.

Table 4 Stage costs of the linked workflow. Conversion and linking are totals per arm; the remaining columns are per requester, with ranges across the reported runs. View checking may build its own index; the later query index is a separate timed stage. Query ranges come from three cold-cache executions after indexing. The whole-genome query index contains 701.5M triples including ClinVar and links.

Arm	RDF	Links	View <i>minutes</i>	Check	Index	Query <i>seconds</i>
Five genomes	3.3	1.4	0.53–0.65	0.35–0.62	2.2	172–181
Cohort	31	22	0.98–1.33	3–5	2.4–3.3	176–202
Whole, clinical	54	8.0	38	68	45	205–209
Whole, empty	—	—	28	1	2.2	< 1

3.3 RQ3: practicality depends on granularity, setup and workload

3.3.1 Sample granularity determines the amount of graph

On the record ladder, triple production was approximately proportional to records at 170.5–171.9 triples per record. Mapping-stage resident memory remained 1.0–1.7 GB; total wall time and disk grew with the data. This supports bounded-memory mapping, not a claim that every downstream validator or query engine is memory-bounded.

The sample ladder shows a more consequential design choice (Figure 2). At 10,000 records and 2,504 samples, expanded produced 627.4M triples and a 4.10 GB HDT, against 59.2 MB of HDT in condensed form. The triple-count ratio was 432 $\times$, but the HDT-byte ratio was 69 $\times$. Condensed triple count grew only 0.9% across the ladder while its HDT grew 5.8 $\times$, because the vectors lengthened. Its lower cost is useful only where applications can accept decoding rather than direct sample-call addressability. Neither profile was used to convert the complete 2,504-sample call set.

3.3.2 Optional artifacts should not be counted as mandatory conversion

Space-optimized storage reduced transient workspace by 9.19 $\times$ on the 100,000-record input at a 5.2% time cost, without changing the triples. In the corpus, COTTAS occupied 0.37–0.55 times the gzip-framed N-Triples and 0.27–0.37 times HDT, but took 1.5–3.0 times longer than HDT to build. These are measured format and implementation choices, not an intrinsic requirement of semantic conversion.

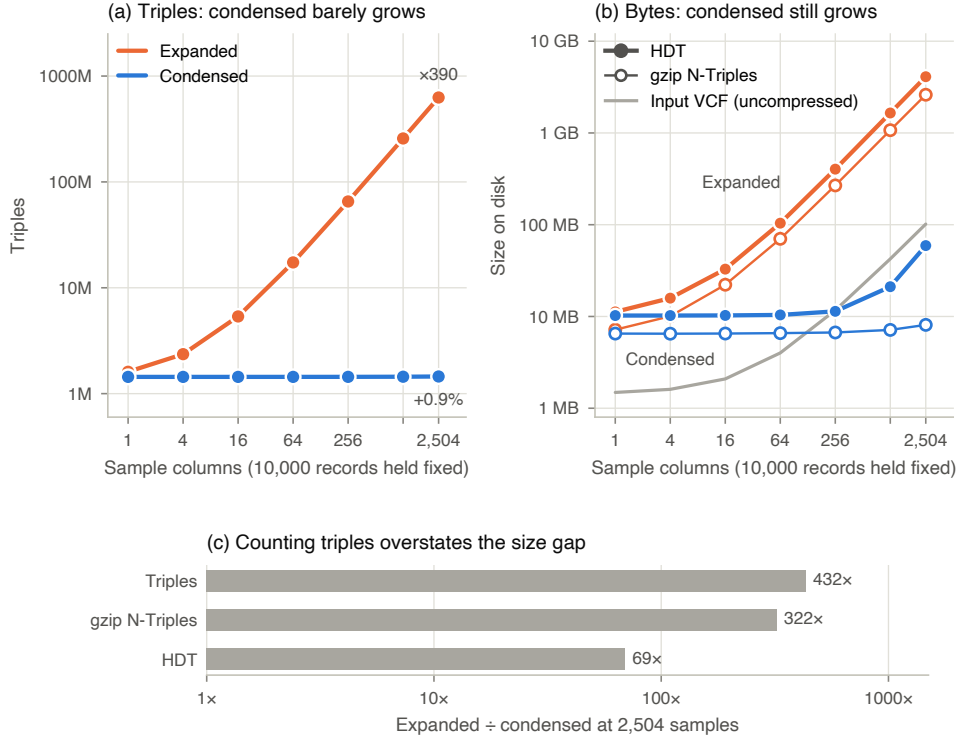


Fig. 2 Sample representation trades addressability for size. The same 10,000 records are emitted with one to 2,504 samples. (a) Expanded resources grow with sample calls; condensed vectors keep triple counts nearly constant. (b) Both representations grow in bytes. (c) At 2,504 samples the expanded/condensed ratio depends on the measure. The $\times 390$ annotation in (a) instead compares expanded output at the two ends of the ladder. Medians are shown where replicated.

The v3.1.0 whole-genome benchmark took 16.07 h including both optional builds: 20,911 s for HDT and 31,230 s for COTTAS, approximately 14.5 h together. Retaining HDT, its index and COTTAS used a reported 10.88 GB; gzip-framed N-Triples alone occupied 2.89 GB. These are alternative artifact choices, not a single minimum storage footprint. They must not be conflated with the later, differently configured 54-minute use-case conversion (with ClinVar). Supplementary Section S6 retains per-input sizes, build times and workspace measurements.

3.3.3 Indexed retrieval benefits some questions, not every workflow

For individual questions on 17.1M triples, indexed QLever took 0.005–4.62 s against cyvcf2 scans of 11.5–12.3 s (Figure 3). Yet QLever first needed a 22.8-second index, in addition to conversion. For the complete thirteen-question batch, one parser pass took 12.4 s, compared with 17.1 s of QLever queries before setup. Thus the tested single cold questions and one-pass summaries favored the parser; selected repeated questions favored the existing graph index.

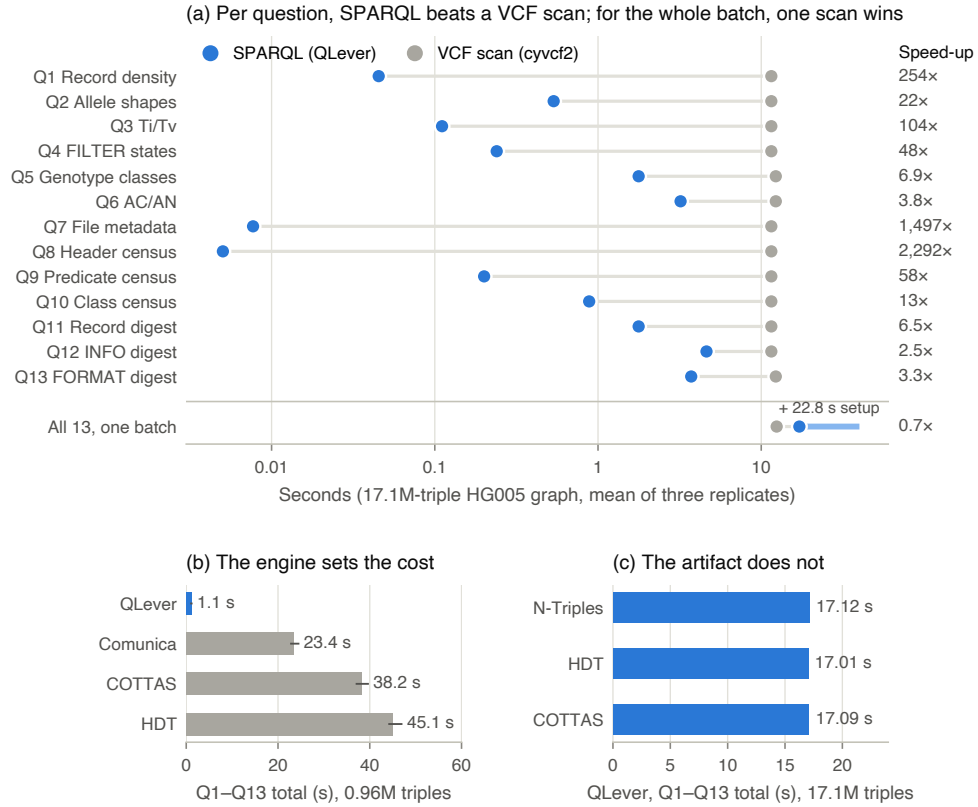


Fig. 3 Retrieval cost depends on the question and execution path. Answers agree before timings are compared. (a) Individual questions and the batch on 17.1M triples, mean of three replicates; conversion is excluded and QLever indexing appears only on the batch row. (b) Four engines on a different, 0.96M-triple graph, excluding setup; means and ranges over nine runs. (c) QLever on 17.1M triples loaded from three artifacts. Similar times in (c) describe QLever’s index, not native querying of those formats.

Regional retrieval provides the stronger indexed-VCF baseline (Table 5). Across both graphs, 11,160 executions had no failures or answer disagreements. Indexed VCF was faster at 1 kb; QLever was faster for the tested 10 Mb windows. Its observed cost was nearly flat across these windows, whereas indexed-reader costs increased with the records processed. Setup remained unequal: 23.2 s for QLever versus 0.48 s for bgzip and tabix.

At whole-genome scale, QLever answered all thirteen inspection questions in 634.6 s after an 868.0-second index build, with every answer equal to the parser’s. These are separate retrieval checks without SHACL. Loading the same graph from HDT or COTTAS changed subsequent query time by only 0.3%, but decoding costs differed. The native HDT- and COTTAS-backed paths did not complete at 171M triples, despite answering five questions. Supplementary Table S13 reports both the

Table 5 Regional retrieval on 17.1M triples, excluding setup. Entries are medians in milliseconds over five questions’ window medians and three repetitions. Indexed arms use twenty windows per size; the unindexed scan uses three. All arms use the same POS-based selection.

Execution path	1 kb	100 kb	1 Mb	10 Mb
QLever	9.6	10.0	9.6	11.1
cyvcf2, indexed	3.3	4.8	14.9	117.8
bcftools, indexed	4.3	4.8	7.5	32.7
cyvcf2, unindexed	635.2	638.0	632.0	649.9

completions and failures; small-fixture engine rankings do not establish large-graph feasibility.

4 Discussion

4.1 The benefit is reusable meaning, not a universally better parser

The successful bcftools comparator is important: the integration task is not exclusive to RDF. What VCF-RDFizer contributes is a common place to state and inspect the relationships on which the task depends. Source observations retain their own context, external classifications and frequencies can be managed separately, and release rules can target declared identities or selections. The frequency join and the identity-targeted policy make this distinction concrete. They demonstrate a reusable implementation of the tested workflow, not that conventional pipelines cannot express it.

Shared identifiers nevertheless require a normalization contract. Minimally trimmed SPDI expressions are not necessarily NCBI’s contextual or canonical alleles in repeat regions [46]. An equivalent allele with a different identifier can miss an equality join and an identity-targeted rule. The demonstration establishes contig-alias handling and agreement after the stated preprocessing; broader identity equivalence remains a boundary to test. Likewise, simulated consent establishes executable rule behavior, not the validity of real consent or protection against access outside the evaluator. Multi-sample masking, enforcement of obligations and secure deployment remain separate work.

4.2 Validation needs to test both the graph and its oracle

The strongest validation result is not simply a high pass count. It is the distinction between what source comparisons see, what structural constraints see, and what neither current implementation sees at scale. The injected faults expose that division, while the real defects show why dependencies matter. The consumer-genome follow-up adds another lesson: a valid conversion can fail when the oracle omits a modeled structure or relies on a different lexical form for an equal value.

The immediate priorities are therefore explicit. Extend census expectations to phase sets and other emitted structures, make value digests insensitive to permitted numeric serialization differences, and implement affordable checks for the dependencies now covered only by deep shapes. Until then, the 984 applicable campaign passes, the separate whole-genome summary agreement and the 58.2M-triple diagnostic comparison must remain distinct claims. No experiment establishes exhaustive semantic equivalence or the biological correctness of the original calls.

4.3 Representation and workload determine the deployment choice

Expanded and condensed output exchange directly addressable structure for compact vectors. That affects queries, validation and policy targeting as well as storage. Condensed sample-level checking needs vector-aware logic, and withholding an individual value would require rewriting the vector; the implemented release demonstration does

neither. Triple counts alone are consequently an inadequate measure of the cost or capability of a genotype representation.

For a one-pass analysis or a narrow indexed lookup, the measured native VCF paths avoid setup that the RDF route cannot recover in that task. For repeated, linked or policy-aware questions, gzip-framed N-Triples and an appropriate graph index are sufficient; optional HDT/COTTAS archives should be built only when their storage or access properties are needed. The 44-question break-even point is one scenario, not a tool constant: it includes a 423.7-second conversion workflow with HDT plus 22.8s of indexing, divided by about 10.2s saved per separate question. It assumes the measured question mix and repeated parser scans. A batched parser pass changes the comparison, and the minimal RDF-only setup was not isolated by that estimate (Supplementary Section S6).

4.4 Scope and next steps

The evidence supports a software and workflow contribution, not clinical deployment, measured usability or superiority over other RDF converters. The converter comparison is documentation-based; the task comparison uses bcftools. Sample scaling holds INFO fixed, the whole cohort was not converted, and full-shape validation was limited to small inputs. The use case has single-sample files and simulated policies. Release-specific results must also remain tied to the versions that produced them.

Focused next steps are more useful than another broad benchmark: resolve the observed oracle failures, complete a conformant sites-only validation, test equivalent variant representations against identity-targeted policies, and assess selective non-empty release at whole-genome scale. A shared-input comparison with another converter would strengthen comparative claims. GA4GH VRS identifiers, federated queries and multi-sample policy evaluation are extensions, not demonstrated capabilities of this study.

5 Conclusions

VCF-RDFizer makes VCF information explicit in a representation that can be checked against its source, connected to external knowledge and used in policy-aware research queries. The evaluation demonstrates complementary validation, reusable joins and requester-specific result agreement with a conventional workflow, while exposing limits in oracle coverage, identity assumptions and large-graph validation. Its measured costs support choosing semantic conversion for tasks that need those relationships and checks, rather than treating RDF as a universal replacement for native VCF processing.

Availability and requirements

Project: VCF-RDFizer. **Home page:** <https://github.com/ecrum19/VCF-RDFizer>. **Language:** Python, with the mapping, compression and query tools specified in Supplementary Section S1. **Platforms:** Linux, macOS and Windows are covered by the tool's tests; the experiments reported here ran on Ubuntu. **Distribution:**

PyPI, Conda-Forge and Docker. **License:** MIT. **Any restrictions to use by non-academics:** none. Docker supplies the pinned benchmark environment; requirements for other installations are documented with each release. v3.1.0 produced the base benchmark, v3.2.0 converted the use-case inputs, and v3.3.0 releases the linking and policy extensions evaluated in pre-release builds. The release ledger distinguishes those builds rather than attributing every result to one version.

Declarations

Ethics approval and consent to participate

This study used publicly released genomic data and involved no recruitment of or interaction with participants. Personal Genome Project donors released their files under the project’s open-consent model [48]; GIAB and precisionFDA files are public reference materials [49, 52]; the 1000 Genomes and HGSC2 datasets are released for research reuse [50, 51, 54]; and ClinVar is a public NCBI resource [55]. The experimental consents are simulated and do not express any real participant’s wishes. No identification of participants was attempted. Ethics approval was not sought for the reported work.

Consent for publication

Not applicable. The manuscript reports aggregate results and software measurements, not individual-level genomic records, images or personal details.

Availability of data and materials

Input filenames, provider URLs and download commands, fixtures, derivation scripts, query files, manifests, normalized outputs and comparison reports are maintained at <https://github.com/ecrum19/vcf-rdfizer-testing>. Large source VCFs are obtained from their public providers rather than redistributed in Git. Experiment 17 (`benchmarks/use_case/acmg`) contains the use-case definitions, simulated policies, conventional baseline and applied-change tests. Its match sets, timings and check reports, and the follow-up validation runs, are recorded under `BioMedSem_2026/benchmark-results`. No biological materials were generated or used.

The software is available at the project home page. The base benchmark used v3.1.0 (`d3b34d5`); the linking and policy extensions are released in v3.3.0 (`7ab2300`; image `sha256:1838720d...06432`); the use case ran pre-release builds (`578c2be` for the whole genome), which v3.3.0 changes only by refusing an empty selector list and an RDF input without VCF triples. Supplementary Section S1 identifies the pre-release and later-run commits, the v3.1.0 image digest, and the bundled shape snapshot. The public vocabulary is separately versioned [28]. Results must be traced to their recorded build and input, not only to the latest package.

[AUTHOR QUERY: Add a persistent archive identifier containing the exact evaluated code, dirty-tree changes for use-case arms 1–2, complete image digests, vocabulary/shapes, queries and evidence manifests. Record the precise commits and configurations for the new default-shape mutation and NG131FQA1I follow-up runs. The supplied manuscript package does not itself establish this complete archived snapshot.]

Competing interests

None declared.

Funding

Elias Crum acknowledges support from the Research Foundation–Flanders (FWO), SB Fellowship 1S27825N.

[AUTHOR QUERY: Complete the funding statements for B.B., R.T. and G.E.]

Authors' contributions

E.C. and G.E. conceived the tool and the experiments. E.C. implemented the workflow and executed the experiments. E.C. and R.T. analyzed the results. E.C. wrote the manuscript draft. E.C., R.T., G.E. and B.B. reviewed and edited the manuscript.

Acknowledgments

Benchmarking used the resources of IDLab at Ghent University. Claude Opus 5.5 [56] and GPT-6 Astra [57] assisted with experiment automation scripting, result data analysis scripting, and restructuring and language editing of this manuscript. The authors reviewed all AI outputs and take full responsibility for the manuscript, experiments, tools, and repositories.

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